Vaccination Cohort & Dose Scheduling System

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Primary Domain: Healthcare & Telemedicine

Target Actors: Citizen Registrant; Vaccination Officer; Notification Service

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Citizen Registrant to create a vaccination registration record using required personal and contact details.	High	A valid registration submission creates exactly one citizen record and displays a registration confirmation.	Registration is required before the citizen can be placed in a vaccination cohort.
FR-002	Functional	The system shall allow a Vaccination Officer to assign a registered citizen to an eligible vaccination cohort based on vaccine and dose eligibility.	High	For an eligible citizen, the officer can select a cohort and the system stores the assignment with date/time and cohort identifier.	Cohort assignment organizes citizens for vaccination delivery.
FR-003	Functional	The system shall allow a Citizen Registrant or Vaccination Officer to schedule an available vaccination dose appointment for an eligible citizen.	High	When an eligible time slot is selected, the system creates one appointment and marks that slot unavailable to other appointments.	Dose scheduling is the core operational function of the system.
FR-004	Functional	The system shall allow a Vaccination Officer to record an administered vaccine dose against a citizen's vaccination record.	High	After the officer submits a valid dose record, the citizen's vaccination history shows the vaccine, dose number, administration date, and officer record.	Accurate dose records are necessary for tracking completed and future doses.
FR-005	Functional	The system shall notify a Citizen Registrant of a confirmed appointment and the next scheduled dose when applicable.	Medium	For every confirmed appointment, a notification is generated containing the appointment date, time, and location; a next-dose notification is generated when a next dose is required.	Reminders reduce missed appointments and support dose completion.
NFR-001	Nonfunctional	The system shall display a successful registration, cohort assignment, or appointment confirmation within 3 seconds under normal operating conditions.	High	In a test of 20 normal transactions, at least 19 complete with the confirmation displayed within 3 seconds.	Fast responses are important for efficient citizen and officer interactions.
NFR-002	Nonfunctional	The system shall restrict vaccination-record updates to authenticated Vaccination Officers and record the identity and timestamp of each update.	High	An unauthenticated user cannot update a vaccination record, and every successful update contains a non-empty officer identifier and timestamp in the audit log.	Vaccination records contain sensitive healthcare information and require controlled modification.

Actors and Use Cases

Citizen Registrant: Registers for vaccination, views eligibility/appointments, and receives scheduling notifications.

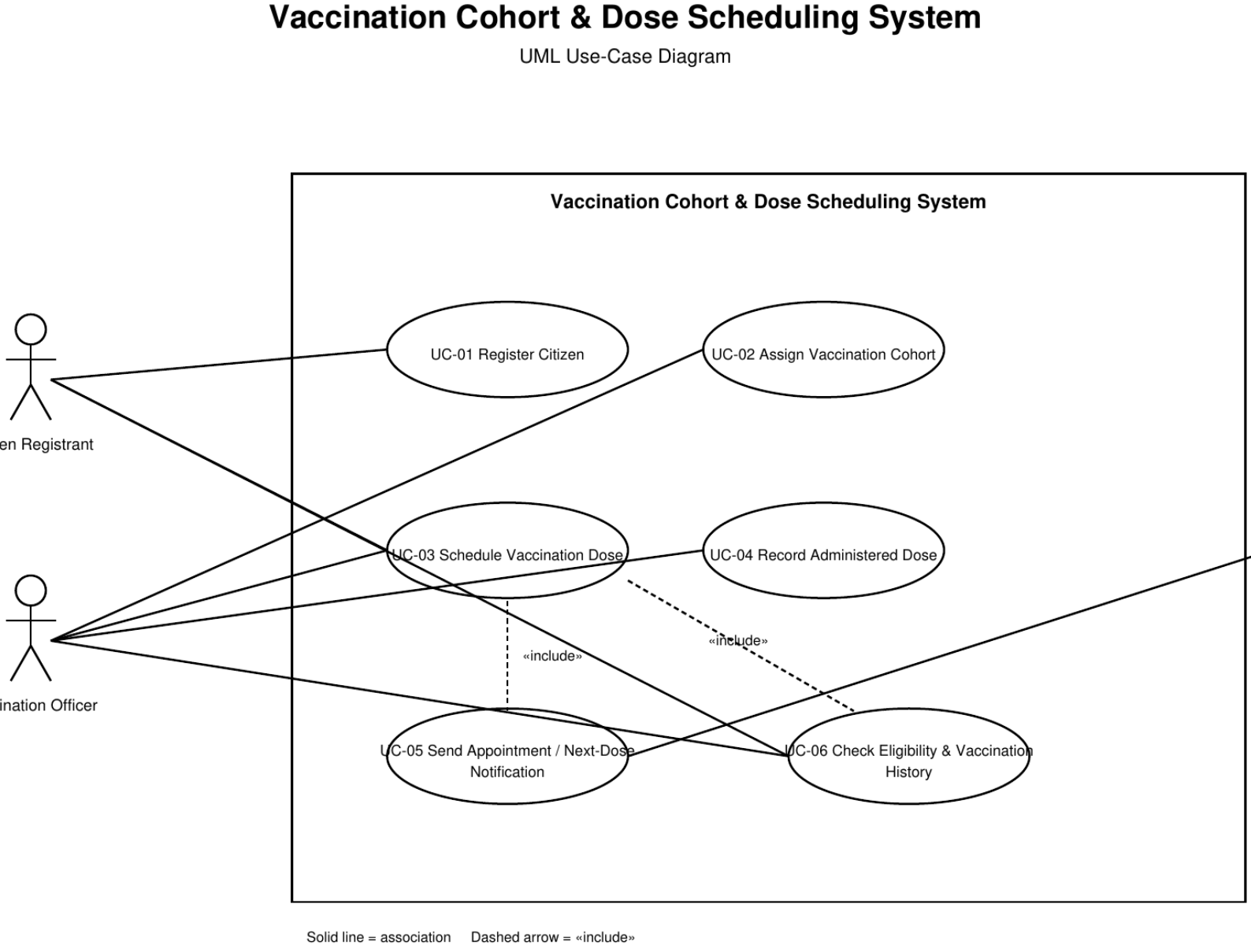
Vaccination Officer: Manages cohorts, schedules doses, and records administered doses.

Notification Service: External service used to deliver appointment and next-dose notifications.

Use Cases: UC-01 Register Citizen; UC-02 Assign Vaccination Cohort; UC-03 Schedule Vaccination Dose; UC-04 Record Administered Dose; UC-05 Send Appointment / Next-Dose Notification; UC-06 Check Eligibility & Vaccination History.

2. UML Use-Case Diagram

Actors, use cases, system boundary, and «include» relationships.



3. Use-Case Flow – Schedule Vaccination Dose

Use-Case Flow – Schedule Vaccination Dose

System: Vaccination Cohort & Dose Scheduling System

Use Case ID: UC-03

Primary Actor: Citizen Registrant

Supporting Actor: Vaccination Officer

Goal: Schedule an available vaccination dose appointment for an eligible citizen.

Preconditions

• The citizen has a valid registration record.

• The citizen is eligible for the selected vaccine/dose.

• At least one vaccination appointment slot is available.

Main Success Scenario

1. Citizen Registrant selects “Schedule Vaccination Dose”.

2. System retrieves the citizen's registration, vaccination history, and eligibility information.

3. System verifies that the citizen is eligible for the required vaccine dose.

4. System displays available vaccination dates, times, locations, and eligible vaccine options.

5. Citizen selects a preferred available appointment slot.

6. System rechecks slot availability and reserves the selected slot.

7. System creates the vaccination appointment and associates it with the citizen and required dose number.

8. System displays the appointment confirmation.

9. System sends the appointment details to the Notification Service for delivery to the citizen.

10. Use case ends successfully.

Alternate Flow – Selected Slot Becomes Unavailable

5a. Citizen selects an appointment slot that was available when displayed.

6a. System finds that the slot has already been booked by another request.

6a1. System does not create an appointment for the unavailable slot.

6a2. System displays a message that the selected slot is no longer available.

6a3. System displays the currently available slots again.

6a4. Citizen selects another available slot and the flow resumes at Step 6.

Postconditions

• A confirmed appointment exists for the citizen, or no appointment is created if the citizen exits without selecting another slot.

The booked slot cannot be assigned to another appointment.

• The citizen receives or is queued to receive the appointment notification.